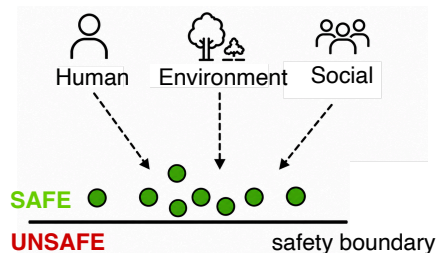


Human-centered Semantic Action Safety

Recurring safety bottlenecks expose four linked requirements for semantic-to-action causality through what must be [1] captured, [2] preserved, [3] enforced, and [4] evidenced.

[1] Safety Boundary Coverage

Data



Curating contextual samples that map the decision boundary for safe next actions.

[2] Semantic-to-Action Fidelity

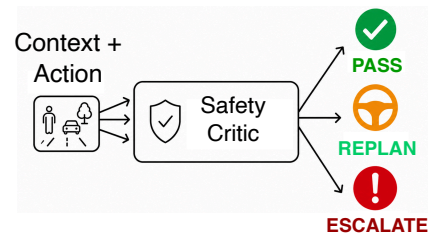
Learning



Reliable semantic action safety requires safety-relevant information to remain causally consequential.

[3] Contextual Intervention & Recovery

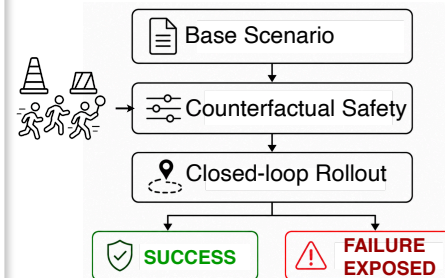
Intervention



Inferring residual errors and uncertainty for context-appropriate intervention and recovery.

[4] Incident-Path Validation

Exposure



Rolling out at safety-critical boundaries to expose potential routes to safety failure.

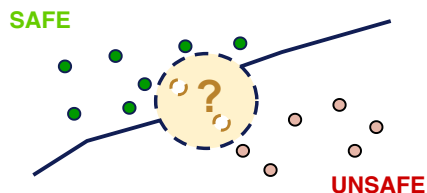
Feedback loop

Open Challenges

However, progress is challenged by [A] sparse boundary examples, [B] limited causal verifiability, [C] intervention timing and recovery gaps, and [D] incomplete incident attribution.

[A] Sparse Boundary Examples

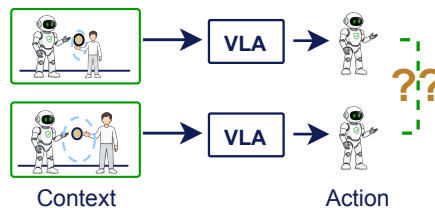
Data



Data collection and generation broaden hazard coverage, but examples distinguishing when the same action is safe or unsafe remain sparse.

[B] Limited Causal Verifiability

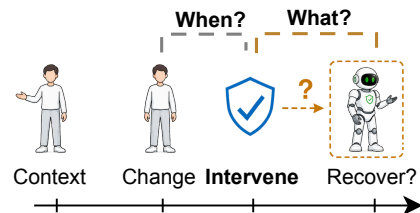
Learning



Constraints, reasoning, and consequence prediction guide safer policies, but the causal influence of safety context on actions remains unclear.

[C] Timing, Recovery Gaps

Intervention



Risk and uncertainty guide runtime checks, but intervention timing and adaptive recovery remain gaps as context evolves.

[D] Incomplete Incident Attribution

Exposure



Benchmarks, stress tests assess safety reasoning and control, but not necessarily the causal links between semantic errors and incidents.